

ARTIFICIAL INTELLIGENCE

“Can Machines Really Think?”

What is AI?

Where do we see AI?

What makes a machine intelligent?

What is an Intelligent Agent?



ARTIFICIAL INTELLIGENCE

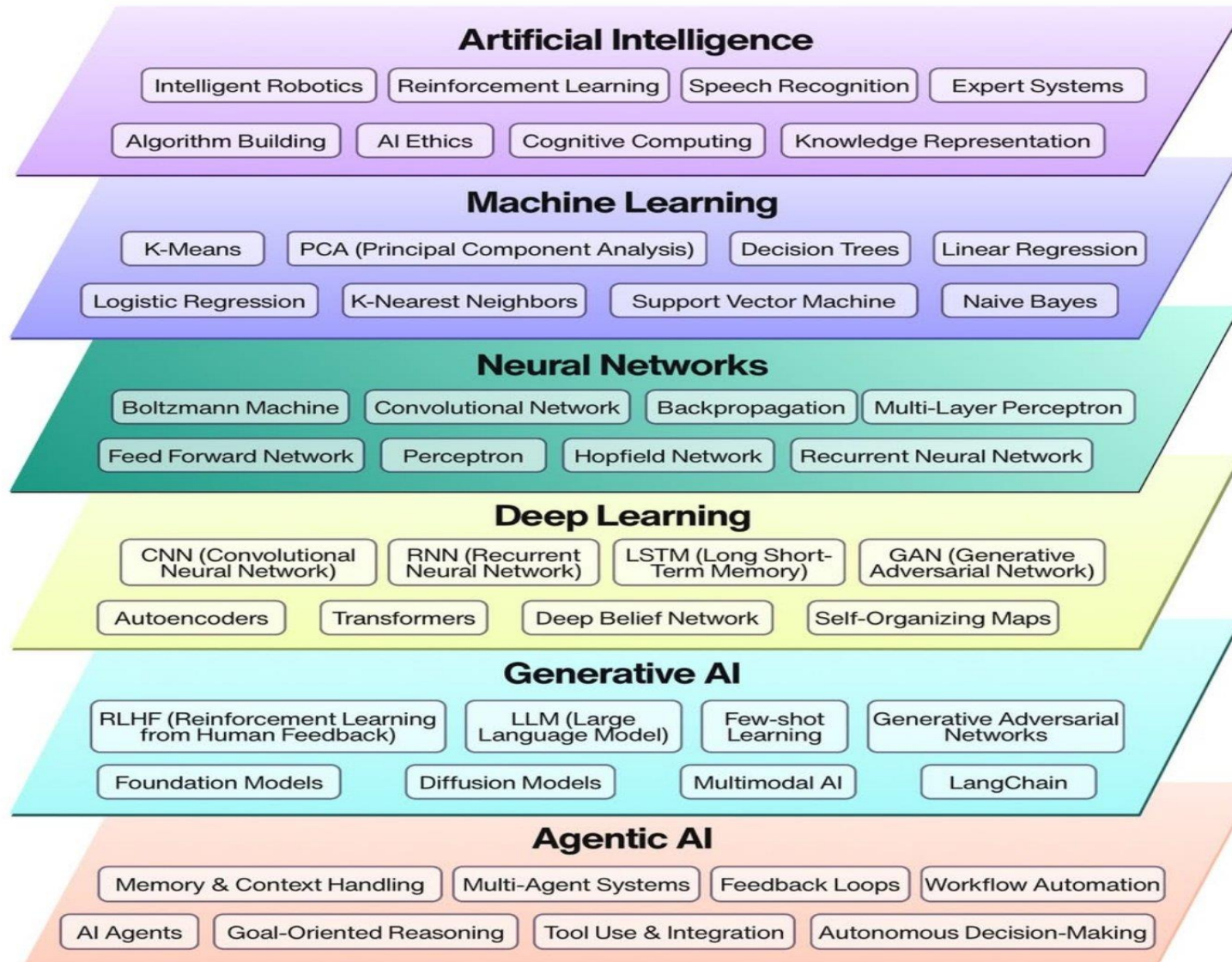
How much AI did you use today?

Think about your morning:

-  Face Unlock
-  Google Maps
-  YouTube / Spotify recommendations
-  Amazon / Flipkart recommendations
-  ChatGPT
-  Google Translate
-  UPI / Banking fraud detection



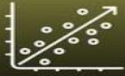
Layers of AI



Top AI Algorithms

And their Use-Cases

Linear Regression



Predicting house prices based on features.

Logistic Regression



Spam email classification.

Decision Trees



Customer churn prediction.

Random Forest



Stock price prediction

K-Means Clustering



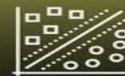
Customer segmentation for marketing.

Naive Bayes



Text classification (e.g., spam or not)

Support Vector Machines (SVM)



Handwriting recognition in digit classification

Neural Networks



Facial recognition.

Gradient Boosting



Credit scoring for loan approval

K-Nearest Neighbors (KNN)



Movie recommendation system

Principal Component Analysis (PCA)



Image compression

Recurrent Neural Networks (RNN)



Sentiment analysis in text

Genetic Algorithms



Optimize supply chain logistics

Long Short-Term Memory (LSTM)



Stock market prediction

Natural Language Processing (NLP)



Chatbot for customer support

Ant Colony Optimization



Traveling salesman problem

Word Embeddings



Improving search engine relevance

Gaussian Mixture Model (GMM)



Anomaly detection in network security

Association Rule Learning



Market basket analysis in retail

Reinforcement Learning



Game playing (e.g., AlphaGo)



Non-Agentic vs Agentic AI vs AI Agent

Non-Agentic AI

DEFINITION:

Basic prompt-response AI. The model takes one input and gives one output with no memory, reasoning, or tool use.

HOW IT WORKS:

- ▶ The user types in a single prompt or question, and the model processes it in isolation.
- ▶ The model generates a direct response immediately, without checking or refining the result.
- ▶ Once the output is delivered, the interaction ends, and the system does not retain context for future tasks.

USE CASES:

- ▶ A founder asks ChatGPT to summarise a 20-page market report into 5 key points before a pitch meeting.
- ▶ A marketer rewrites a sales email into three different tones to test which performs best.
- ▶ A content creator uses it to generate 20 headline variations for a LinkedIn post in under a minute.

PROS:

- ▶ Fast, cheap, and universally accessible
- ▶ Requires no technical build or integration
- ▶ Great for one-off, clearly defined tasks

CONS:

- ▶ No reasoning or context awareness
- ▶ Output quality depends heavily on prompts
- ▶ Struggles with multi-step or complex work

HOW TO BUILD FAST:

- ▶ Open a tool like ChatGPT, Claude, or Gemini and create a new chat.
- ▶ Write a clear, specific prompt that includes all necessary context upfront.
- ▶ Copy the output into your workflow, edit as needed, and repeat the process for new tasks.

Agentic AI

DEFINITION:

A self-managing AI system that can plan, execute, critique, and iterate across multiple steps.

HOW IT WORKS:

- ▶ The user sets a broad goal, and the system automatically breaks it into smaller sub-tasks.
- ▶ It connects to tools, APIs, or data sources to execute each step, pulling in relevant information as required.
- ▶ Results are evaluated against the original goal, with feedback loops and memory layers enabling continuous improvement.

USE CASES:

- ▶ A consultant asks the system to research 10 competitors, draft a market analysis, and design a slide deck for client delivery.
- ▶ A growth team sets it to manage ad campaigns across Google and Meta, monitoring performance and adjusting spend daily.
- ▶ A SaaS founder uses it to generate onboarding flows, test them with dummy data, and refine copy based on user behaviour.

PROS:

- ▶ Handles complex, changing projects with many moving parts
- ▶ Can integrate with tools, databases, and long-term memory
- ▶ Produces more reliable outcomes than isolated prompts

CONS:

- ▶ Slower and more expensive than simple agents
- ▶ Still requires human oversight and guardrails
- ▶ Risk of overbuilding for simple problems

HOW TO BUILD FAST:

- ▶ Use frameworks like LangChain, CrewAI, or AutoGen to connect models with tools and workflows.
- ▶ Assign roles such as Planner, Executor, and Critic so tasks can be broken down and reviewed automatically.
- ▶ Add memory and feedback layers to prevent the system from resetting each time and to increase consistency.

AI Agent

DEFINITION:

A single-task AI worker designed to automate one repetitive job with speed and reliability.

HOW IT WORKS:

- ▶ The user defines one clear responsibility, such as "update CRM records" or "generate weekly reports."
- ▶ The agent receives inputs, accesses one or two connected tools like calendars, emails, or databases, and executes.
- ▶ Once the task is complete, the agent outputs the result without needing step-by-step human guidance.

USE CASES:

- ▶ A sales leader deploys an agent to pull pipeline data from HubSpot every Friday and update a Google Sheet for the team.
- ▶ A customer service team uses an agent to draft personalised replies to FAQs and send them through email automatically.
- ▶ A finance lead sets up an agent to generate a weekly expense report by pulling transactions from accounting software.

PROS:

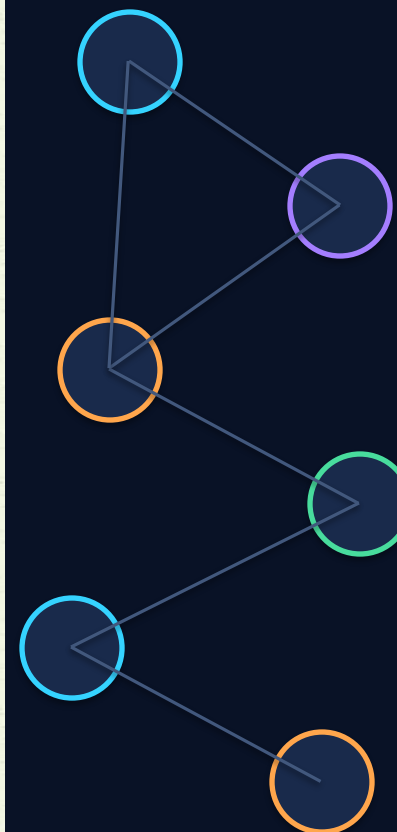
- ▶ Automates repetitive, time-consuming tasks
- ▶ Positions brand as a trusted knowledge source.
- ▶ Easy to test and refine within specific roles

CONS:

- ▶ Limited flexibility - can't handle broader projects
- ▶ Breaks if inputs are unclear or data is missing
- ▶ Needs orchestration to work alongside other agents

HOW TO BUILD FAST:

- ▶ Define one specific, repetitive task where automation would save time or money.
- ▶ Wrap an LLM with tools using Zapier, LangChain, or the OpenAI Assistants API to execute that task.
- ▶ Connect it to the required inputs and outputs (e.g. CRM, email, calendar) and test thoroughly before deploying.



ARTIFICIAL INTELLIGENCE

From “What is AI?” to Problem-Solving Agents

Interactive Lecture Deck

Based on Russell & Norvig, A Modern Approach, 4th ed.



Today's journey: from intelligence to action

AI

01

Define intelligence
and rationality

AGENTS

02

Perceive → decide →
act

ENVIRONMENTS

03

Understand the
world around the
agent

AGENT DESIGN

04

Choose the right
agent architecture

PROBLEM SOLVING

05

Formulate → search
→ execute

THINK → By the end, can you design the *first version* of an AI agent for a real engineering problem?

Warm-up: Is a calculator an AI agent?

CALCULATOR

Input: $2 + 2$

Output: 4

It reacts to input, but does that make it “intelligent”?

AI AGENT

Perceives a situation

Chooses an action

Acts in an environment

Decision-making matters.

ENGINEERING QUESTION

What is the *task*?

What counts as success?

What can the system sense?

What can it do?

THINK → What changes if the environment is uncertain, dynamic, or partially observable?

What is AI?

AI is about building intelligent entities—machines that can compute how to act effectively and safely in a wide variety of novel situations.

PERCEIVE

Understand signals
from the world

REASON

Infer, compare,
and predict

LEARN

Improve from
experience

ACT

Choose actions that
move toward objectives

AIMA framing: AI spans learning, reasoning, perception and action—from games to driving to diagnosis.

Four ways people have defined AI

Human-like

Thought

Rational

THINKING HUMANLY

Cognitive modeling
Psychology + AI

THINKING RATIONALLY

“Laws of thought”
Logic + formal reasoning

ACTING HUMANLY

Turing-test approach
Human-like behavior

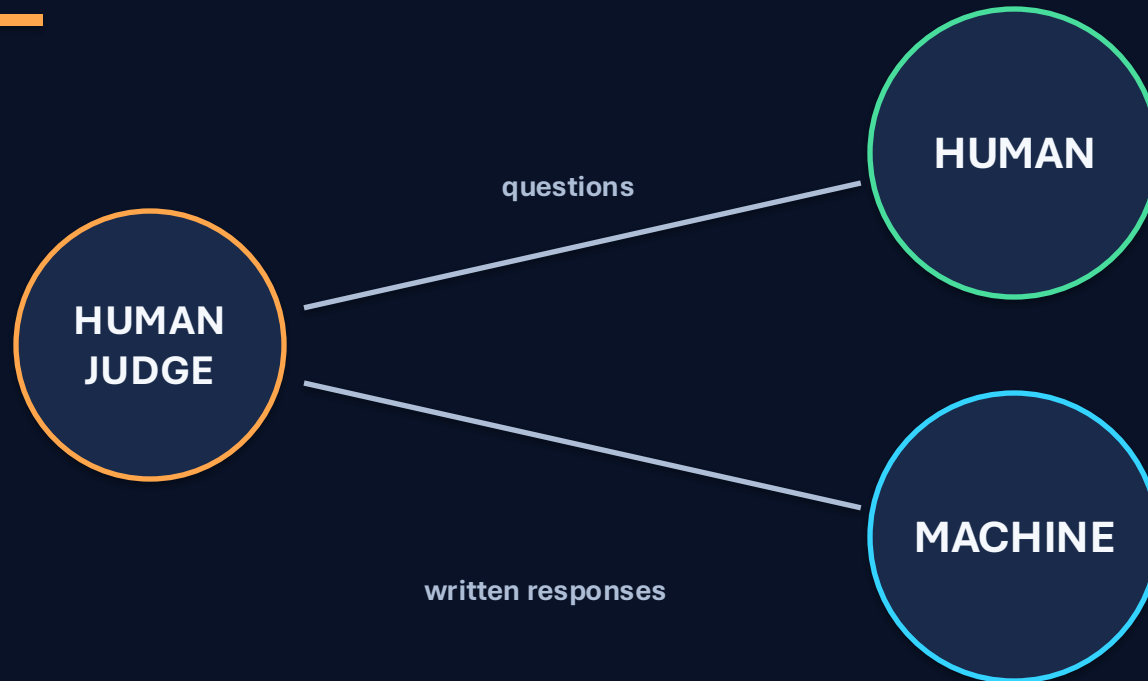
ACTING RATIONALLY

Rational-agent approach
Choose good actions

Behavior

THINK → Which quadrant is most useful if your goal is to engineer a reliable system?

The Turing Test: can behavior look human?



A computer passes if a human interrogator cannot distinguish its written responses from a person's.

The textbook notes capabilities such as natural language processing, knowledge representation, automated reasoning and learning.

A total Turing test adds perception and physical interaction: computer vision, speech and robotics.

THINK → Does human-like behavior necessarily mean the system uses human-like reasoning?

Why the rational-agent approach?

Rationality ≠ “always perfect.”
A rational agent chooses the action expected to maximize performance given what it knows and has perceived.

Engineering lens Specify a performance criterion → design mechanisms that select actions that do well under that criterion.

INFERENCE

Can help choose good actions

REFLEX

Can be rational without deliberation

LEARNING

Improves behavior in new situations

ACTION

Focus: what the agent should do

THINK → If the objective is wrong, can a perfectly rational agent still do the wrong thing?

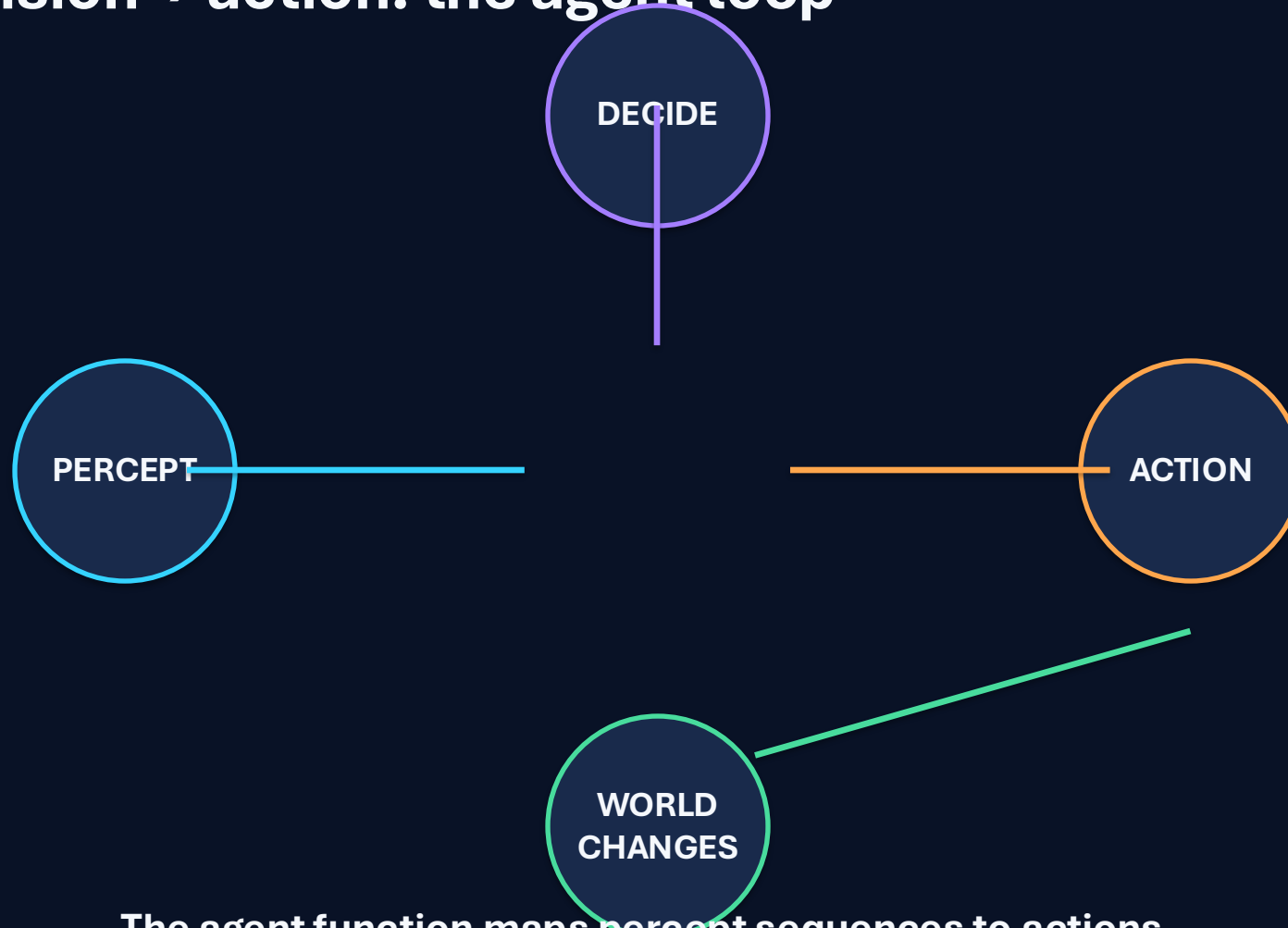
AI's central abstraction: the intelligent agent

An agent perceives its environment through sensors and acts upon it through actuators.



Examples: human • robot • software agent

Percept → decision → action: the agent loop



The agent function maps percept sequences to actions.

Agent function: $f(\text{percept sequence}) \rightarrow \text{action}$

Three terms students must not mix up

PERCEPT

What the sensors are perceiving
now.

Example:

[location = A, status = Dirty]

PERCEPT SEQUENCE

The complete history of what the
agent has perceived.

Example:

[A,Clean] → [A,Dirty] → ...

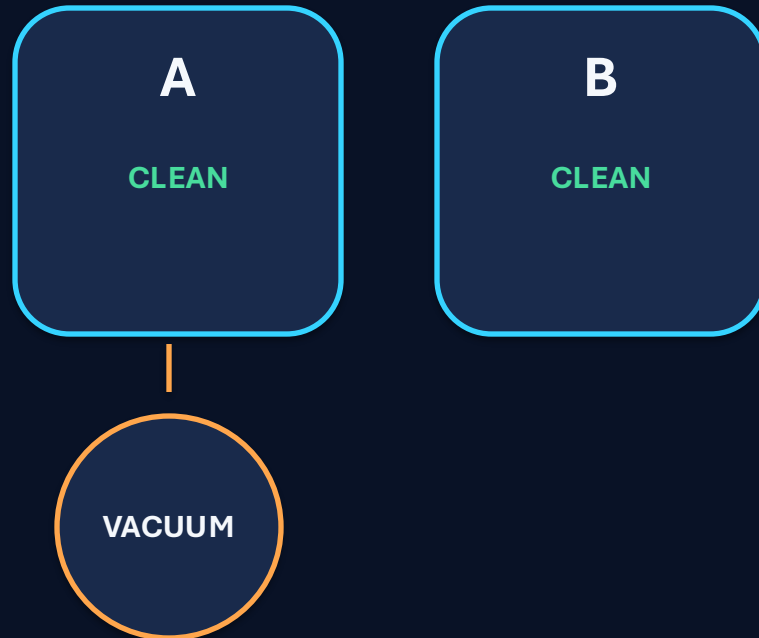
AGENT FUNCTION

A mapping from every possible
percept sequence to an action.

Implementation ≠ function.

THINK → Why might a useful agent need memory if the current percept is incomplete?

Vacuum world: our first AI laboratory



Percepts: current location + whether the square is dirty

Actions: Left, Right, Suck, Do Nothing

A simple rule: if dirty → Suck; otherwise → move to the other square

THINK → Would the same rule still be rational if movement had an energy penalty?

Rationality depends on FOUR things

Performance measure

1

What counts as success?

Prior knowledge

2

What does the agent already know?

Available actions

3

What can it actually do?

Percept sequence

4

What evidence has it received?

Rational choice = best expected performance, given the evidence available.

Rational $\neq$ omniscient $\neq$ perfect

OMNISCIENT

Knows the actual outcome of actions.

Not achievable in reality.

RATIONAL

Maximizes expected performance using available evidence.

PERFECT

Would maximize actual outcome after everything is known.

Not a feasible design target.

THINK → If a rational driver makes a safe decision and an unexpected event causes failure, was the decision irrational?

PEAS: start with the task environment

P**PERFORMANCE**

How do we score success?

E**ENVIRONMENT**

What world does it operate in?

A**ACTUATORS**

What actions can it take?

S**SENSORS**

What can it perceive?

AIMA's design rule: specify the task environment as fully as possible before designing the agent.

PEAS example: automated taxi

Performance

Safe, fast, legal, comfortable; profit; low impact on other road users

Environment

Roads, traffic, pedestrians, police, customers, weather

Actuators

Steering, accelerator, brake, signal, horn, display, speech

Sensors

Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

THINK → Which PEAS item becomes hardest to specify when human preferences are involved?

The nature of environments changes the design

Observable

Fully ↔ Partially
Can I see all relevant state?

Agents

Single ↔ Multi
Are other decision-makers involved?

Outcome

Deterministic ↔ Nondeterministic
Is the next state predictable?

Time

Episodic ↔ Sequential
Do current actions affect the future?

World

Static ↔ Dynamic
Can the world change while I deliberate?

Representation

Discrete ↔ Continuous
States/actions/time: finite or continuous?

Knowledge

Known ↔ Unknown
Do I know the environment's rules?

Key insight: the harder the environment, the more demanding the agent design.

Classify these environments

Chess with a clock

Fully • Multi • Deterministic • Sequential • Semidynamic • Discrete

Poker

Partially • Multi • Stochastic • Sequential • Static • Discrete

Taxi driving

Partially • Multi • Stochastic • Sequential • Dynamic • Continuous

Image analysis

Fully • Single • Deterministic • Episodic • Semidynamic • Continuous

THINK → Which one is closest to the environment of an autonomous delivery robot on a college campus?

Agent = Architecture + Program



ARCHITECTURE

The physical/computing substrate that receives percepts, runs the program, and drives actuators.

THINK → A robot arm has an excellent controller but weak sensors. Can the program alone fix the problem?

Four core agent program designs

Simple reflex

Current percept → rule → action

Fast • compact • limited

Model-based reflex

Percept + internal state → rule → action

Handles partial observability

Goal-based

State + goals + model → future-oriented action

Search & planning

Utility-based

State + utility + model → best expected outcome

Trade-offs under uncertainty

All four can be extended with learning.

Simple reflex agent: “IF this, THEN that”



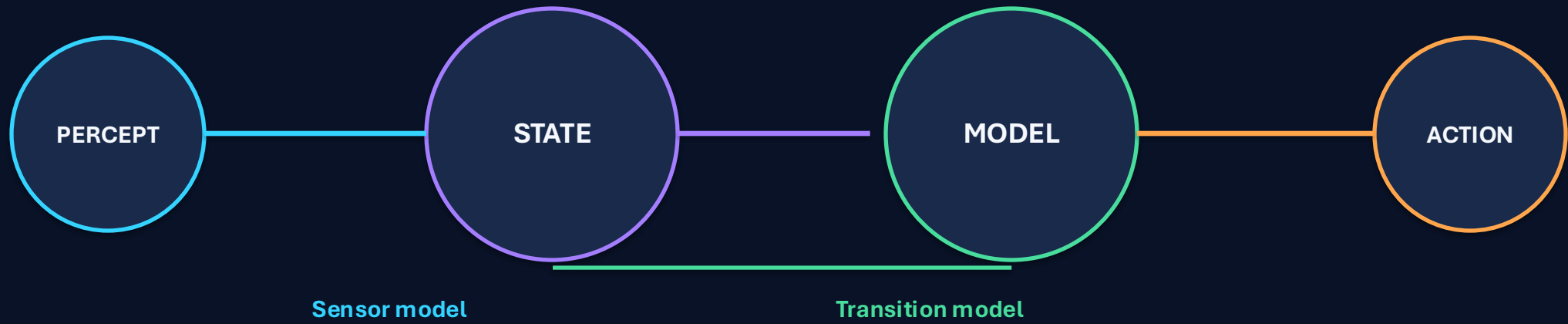
**IF car-in-front-is-braking
THEN initiate-braking**

Strength: simplicity

Weakness: depends heavily on the current percept and works best when relevant state is observable.

THINK → What happens when two different world states look identical to the sensor?

Model-based reflex: remember what you cannot see



Internal state = best available description of relevant hidden aspects of the world.

Goal-based agent: look ahead



Utility-based agent: goals may conflict



Choose the action with the best expected utility.

Utility lets an agent weigh trade-offs among desirable outcomes when certainty is impossible.

Learning agent: improve the agent over time



Any of the agent types can be built as learning agents.

THINK → Why is learning especially valuable when the environment is initially unknown or changes over time?

Problem-solving agent: when the next action is not obvious

A problem-solving agent looks ahead and searches for a sequence of actions that reaches a goal state.



Assumption in this chapter: simple environments that are episodic, single-agent, fully observable, deterministic, static, discrete and known.

Romania route problem: turn the world into a search problem



The map is a model of the relevant part of the world.

Problem formulation: define the pieces

State space

Set of possible states
Where can the agent be?

Initial state

Where the agent starts
Example: Arad

Goal test

Which states count as success?
Example: Is city = Bucharest?

Actions

What can be done in a state?
Example: drive to adjacent city

Transition model

What state results from an action?
 $\text{RESULT}(s, a)$

Action cost

Numeric cost of an action
Distance, time, energy, ...

A sequence of actions forms a path; a solution is a path from the initial state to a goal state.

Search before acting: think in a model first

REAL WORLD

Roads • traffic • pedestrians

Actions change the real environment.

Risk/cost may be high.

SEARCH

INTERNAL MODEL

States • actions • transitions

Simulate candidate action sequences.

Search for a solution.

In the simple setting, once a solution is found, it can be executed as a fixed sequence if the model is correct and the world is deterministic.

Open-loop vs. closed-loop execution

OPEN LOOP

Plan → execute

Works when the environment is fully observable, deterministic and the model is correct.

The agent can ignore new percepts while executing.

CLOSED LOOP

Plan → act → perceive → re-plan

Safer when the model may be wrong or outcomes are uncertain.

Contingencies may be needed.

THINK → For an autonomous robot moving through a crowded corridor, which loop would you trust?

A problem-solving agent in one picture



The key engineering move: reduce a messy world to the **right abstraction of states, actions and goals.**

Good problem formulation → smaller, clearer search space → better chance of a useful solution.

Mini design challenge: build an AI agent

Scenario: Smart campus energy controller

Goal: reduce energy use without compromising comfort and safety.

Identify the PEAS description.

Classify the environment on at least 4 dimensions.

Choose: reflex, model-based, goal-based, utility-based or learning agent.

If it must reach a target state, formulate a search problem.

TEAM OUTPUT

1 minute pitch:

“ Our agent senses ____,
acts using ____,
and optimizes ____.”

THINK → Which design choice is most important—and why?

Exit ticket: 6 questions

1. Define a rational agent in your own words.

2. Distinguish percept from percept sequence.

3. Write PEAS for a campus delivery robot.

4. Give one example of a partially observable environment.

5. Why is a goal-based agent more flexible than a simple reflex agent?

6. Convert “reach lab safely” into a search problem.

Use these as a 10-minute classroom quiz, LMS quiz, or viva prompts.

Key takeaways

AI

Engineering intelligent behavior—not merely computation.

Agent

Perceives + acts; agent function maps percept sequences to actions.

Rationality

Choose actions that maximize expected performance given evidence.

PEAS

Performance • Environment • Actuators • Sensors.

Environment

Its properties determine what agent design makes sense.

Problem solving

Goal → formulation → search → solution → execution.

The unifying idea from ALMA: design the *whole agent* around the task environment.

Reference & teaching note

Primary reference

Stuart Russell & Peter Norvig
Artificial Intelligence: A Modern Approach, 4th
Edition
Pearson

Content basis used in this deck

Chapter 1: Introduction — What Is AI?

Chapter 2: Intelligent Agents — Agents & Environments, Rationality, Nature of Environments, Structure of Agents

Chapter 3, §3.1: Problem-Solving Agents

FACULTY TIP

Use the diagrams as talking points.

Pause after each “THINK →” prompt.

For a 60-minute class: omit Slides 31–32 or use them as tutorial work.